

33.6 Summary and Notes

33.6.1 Summary

Idea.

1. A split-plot design randomizes twice and has two error terms: the response space splits into a whole-plot stratum with error variance $\sigma_s^2 + b\sigma_w^2$ and a subplot stratum with error variance σ_s^2 (Lemma 33.1.1).
2. In the balanced design ordinary least squares is still best, because $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$. The whole-plot factor is tested against the block-by-treatment mean square on $(r-1)(a-1)$ degrees of freedom, the subplot factor and the interaction against the residual; a comparison of whole-plot treatments at one subplot level belongs to neither stratum and needs both mean squares (Theorem 33.1.2). Pooling inflates the whole-plot F and deflates the subplot one.
3. With compound symmetry inside clusters of size m , a cluster-level coefficient has its variance multiplied by the design effect $1 + (m-1)\rho$ and a within-cluster coefficient by $1 - \rho$; the usual standard error is wrong by $\{E(s^2)/\lambda\}^{1/2}$, with $E(s^2)$ given exactly (Proposition 33.2.1). Clustering moves information from between-cluster to within-cluster comparisons.
4. The cluster-robust covariance estimator assumes only that clusters are independent, and is consistent as the number of clusters grows, not the number of observations. With few clusters it is badly biased downward.
5. The repeated-measures F test is exact if and only if the covariance satisfies sphericity: all pairwise differences of the repeated measurements have the same variance (Lemma 33.3.1, Theorem 33.3.2). Compound symmetry implies sphericity; the converse is false.
6. When sphericity fails, the moment-matched approximation multiplies both degrees of freedom by Box's ϵ (Proposition 33.3.3); the Greenhouse–Geisser and Huynh–Feldt corrections estimate ϵ , and Hotelling's T^2 avoids the issue at a cost in power (Proposition 33.3.4).
7. Longitudinal covariance is built from random effects, serial correlation and measurement error, and the standard models keep one or two of the three (Definition 33.4.1). A wrong covariance model leaves the fixed effects unbiased and is repaired by the empirical standard error; restricted likelihoods compare only models with the same mean structure (Proposition 33.4.1).
8. In the balanced random-coefficient growth model the population curve is the average of the subjects' own curves, needing no covariance parameter; the predicted subject curve shrinks the subject's own curve towards it by $\mathbf{D}(\mathbf{D} + \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})^{-1}$; and restricted maximum likelihood has the closed form of the two-stage moment estimator (Proposition 33.5.1).

33.6.2 Notes and sources

Split plots. The design and its two-stratum analysis are due to Yates (1935); Fisher (1935) had already made the choice of error term the central issue of experimental design. The route

taken here is Lemma 33.1.1, which reduces everything to Theorem 16.3.1 plus Kruskal's condition (Theorem 6.9.3); Christensen (2020, chapter 11) develops split-plot models from a cluster sampling model in the same coordinate-free spirit. The cross-stratum simple effect of Theorem 33.1.2(f) is standard practical advice given a derivation here.

Clusters and the design effect. The design effect $1 + (m - 1)\rho$ is Kish's (1965), from survey sampling; its damage to standard errors for aggregate regressors was pressed on econometricians by Moulton (1990). The cluster-robust estimator Eq. 33.2.3 is the clustered form of the sandwich estimators of Section 21.4, whose proof also gives its consistency, cluster by cluster in place of observation by observation; its small-sample failings and their remedies are surveyed by Cameron and Miller (2015).

Sphericity. Box (1954) computed the effect of an arbitrary covariance on the two-way analysis of variance and introduced ϵ and the approximation of Proposition 33.3.3. Huynh and Feldt (1970) isolated the necessary and sufficient condition of Lemma 33.3.1, which is named after them, and the fixed- N form of Theorem 33.3.2(c) is theirs; the proof here takes the easier route through the large-sample limit. Geisser and Greenhouse (1958) showed the lower-bound reference distribution to be exactly conservative, not merely so within Box's approximation; Greenhouse and Geisser (1959) made the plug-in and lower-bound corrections a recommendation, Huynh and Feldt (1976) gave the bias-corrected Eq. 33.3.3, and Mauchly (1940) the likelihood ratio test of sphericity itself. The multivariate alternative is Hotelling's T^2 (Hotelling 1931), whose distribution theory we quote from Anderson (2003).

Longitudinal covariance. The three-source decomposition Eq. 33.4.1 and the empirical variogram are due to Diggle (1988); Diggle, Heagerty, Liang and Zeger (2002) is the standard account. That a working covariance can be wrong without invalidating the estimate, provided the standard error is computed empirically, is from Liang and Zeger (1986), whose generalized estimating equations are the subject of Chapter 40. The rule of Proposition 33.4.1(c) follows from Definition 32.4.1 and is a common source of error in practice. Degrees of freedom under an estimated covariance belong to Satterthwaite (1946) and Kenward and Roger (1997); missing data to Chapter 41, the missing-at-random condition being Rubin's (1976).

Growth curves. Potthoff and Roy (1964) introduced Eq. 33.5.1 and reduced it to an ordinary multivariate analysis of variance by a weight matrix fixed in advance; Rao (1965) gave the efficient estimate and recast growth curves in terms of random coefficients, Eq. 33.5.2; Laird and Ware (1982) put the two together in the form that became standard software. The shrinkage Eq. 33.5.5 is the multivariate signal-to-noise weighting of Chapter 27; the closed form Eq. 33.5.7 appears to be folklore, proved here by factorizing the error-contrast likelihood into within- and between-subject pieces.

The data. The investment panel of Example (Investment in eleven firms) and Example (The investment trajectories of eleven firms) is Grunfeld's (1958), in the eleven-firm version reconstructed by Kleiber and Zeileis (2010) and distributed with statsmodels — the panel already fitted with a random firm effect in Chapter 32, seen here from two more angles. The split-plot and repeated-measures examples are simulated.

33.6.3 References

- Anderson, Theodore W. (2003). *An Introduction to Multivariate Statistical Analysis*. 3rd edition. Hoboken, NJ: Wiley.
- Box, George E. P. (1954). Some Theorems on Quadratic Forms Applied in the Study of Analysis of Variance Problems, II. Effects of Inequality of Variance and of Correlation

- Between Errors in the Two-Way Classification. *The Annals of Mathematical Statistics* 25(3), 484–498.
- Cameron, A. Colin and Miller, Douglas L. (2015). A Practitioner’s Guide to Cluster-Robust Inference. *Journal of Human Resources* 50(2), 317–372.
 - Christensen, Ronald (2020). *Plane Answers to Complex Questions: The Theory of Linear Models*. 5th edition. Cham: Springer.
 - Diggle, Peter J. (1988). An Approach to the Analysis of Repeated Measurements. *Biometrics* 44(4), 959–971.
 - Diggle, Peter J., Heagerty, Patrick, Liang, Kung-Yee and Zeger, Scott L. (2002). *Analysis of Longitudinal Data*. 2nd edition. Oxford: Oxford University Press.
 - Fisher, Ronald A. (1935). *The Design of Experiments*. Edinburgh: Oliver and Boyd.
 - Geisser, Seymour and Greenhouse, Samuel W. (1958). An Extension of Box’s Results on the Use of the F Distribution in Multivariate Analysis. *The Annals of Mathematical Statistics* 29(3), 885–891.
 - Greenhouse, Samuel W. and Geisser, Seymour (1959). On Methods in the Analysis of Profile Data. *Psychometrika* 24(2), 95–112.
 - Grunfeld, Yehuda (1958). *The Determinants of Corporate Investment*. PhD thesis, University of Chicago.
 - Hotelling, Harold (1931). The Generalization of Student’s Ratio. *The Annals of Mathematical Statistics* 2(3), 360–378.
 - Huynh, Huynh and Feldt, Leonard S. (1970). Conditions under Which Mean Square Ratios in Repeated Measurements Designs Have Exact F -Distributions. *Journal of the American Statistical Association* 65(332), 1582–1589.
 - Huynh, Huynh and Feldt, Leonard S. (1976). Estimation of the Box Correction for Degrees of Freedom from Sample Data in Randomized Block and Split-Plot Designs. *Journal of Educational Statistics* 1(1), 69–82.
 - Kenward, Michael G. and Roger, James H. (1997). Small Sample Inference for Fixed Effects from Restricted Maximum Likelihood. *Biometrics* 53(3), 983–997.
 - Kish, Leslie (1965). *Survey Sampling*. New York: Wiley.
 - Kleiber, Christian and Zeileis, Achim (2010). The Grunfeld Data at 50. *German Economic Review* 11(4), 404–417.
 - Laird, Nan M. and Ware, James H. (1982). Random-Effects Models for Longitudinal Data. *Biometrics* 38(4), 963–974.
 - Liang, Kung-Yee and Zeger, Scott L. (1986). Longitudinal Data Analysis Using Generalized Linear Models. *Biometrika* 73(1), 13–22.
 - Mauchly, John W. (1940). Significance Test for Sphericity of a Normal n -Variate Distribution. *The Annals of Mathematical Statistics* 11(2), 204–209.
 - Moulton, Brent R. (1990). An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units. *The Review of Economics and Statistics* 72(2), 334–338.
 - Potthoff, Richard F. and Roy, S. N. (1964). A Generalized Multivariate Analysis of Variance Model Useful Especially for Growth Curve Problems. *Biometrika* 51(3–4), 313–326.
 - Rao, C. Radhakrishna (1965). The Theory of Least Squares When the Parameters Are Stochastic and Its Application to the Analysis of Growth Curves. *Biometrika* 52(3–4), 447–458.
 - Rubin, Donald B. (1976). Inference and Missing Data. *Biometrika* 63(3), 581–592.
 - Satterthwaite, Franklin E. (1946). An Approximate Distribution of Estimates of Variance Components. *Biometrics Bulletin* 2(6), 110–114.
 - Yates, Frank (1935). Complex Experiments. *Supplement to the Journal of the Royal Statistical Society* 2(2), 181–247.